

PRINCE RUPERT, BC, Canada

WMO: 710220

Lat: 54.290N

Lon: 130.440W

Elev: 35

StdP: 100.90

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.3	-17.5	0.8	-5.5	-13.3	1.2	-4.3	14.2	7.2	12.8	6.6	2.4	70	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	19.6	14.9	18.0	14.4	16.9	14.0	15.9	18.2	15.2	17.1	14.5	16.2	3.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	10.5	16.6	14.2	10.1	16.0	13.6	9.8	15.4	44.4	18.4	42.4	17.2	40.7	16.3	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.2	9.4	8.0	DB	-11.7	24.0	3.6	2.6	-14.3	25.9	-16.4	27.4	-18.4	28.9	-21.1	30.8
(5)				WB	-12.4	17.6	3.5	0.9	-15.0	18.2	-17.0	18.8	-19.0	19.3	-21.5	19.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	2.8	3.0	4.3	6.6	9.5	12.0	13.8	14.2	12.2	8.6	4.8	2.9
	DBStd	4.98	4.18	3.23	2.83	2.04	2.18	1.80	1.51	1.46	1.91	2.21	3.28	3.74
	HDD10.0	1172	222	197	178	103	34	2	0	0	4	53	156	222
	HDD18.3	3806	481	431	437	351	273	191	141	130	186	302	405	480
	CDD10.0	409	0	0	0	2	20	62	117	129	68	9	1	0
	CDD18.3	1	0	0	0	0	0	1	0	1	0	0	0	0
	CDH23.3	9	0	0	0	0	1	6	1	2	1	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	3.3	3.7	3.6	3.9	3.6	3.1	2.8	2.5	2.6	3.0	3.7	3.9	4.0
Precipitation	PrecAvg	2690	259	223	192	182	145	123	118	170	253	375	299	293
	PrecMax	3282	437	398	312	337	335	242	240	541	515	748	601	566
	PrecMin	2297	96	6	38	60	62	30	15	30	79	163	94	96
	PrecStd	265	81	93	63	71	63	53	60	92	109	118	115	115
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	11.2	14.1	17.0	20.9	22.9	21.4	22.1	20.1	15.9	12.7	11.6
		MCWB	8.0	6.7	8.6	9.2	12.1	15.7	16.6	16.0	15.5	11.3	8.9	7.6
	2%	DB	9.4	9.7	10.7	13.5	16.4	17.7	18.5	19.0	17.6	14.0	11.1	9.7
		MCWB	7.6	6.5	7.0	8.4	11.4	13.8	15.2	15.3	13.9	11.1	8.9	7.2
	5%	DB	8.5	8.4	9.1	11.3	14.2	16.0	17.3	17.8	16.2	12.7	10.0	8.6
		MCWB	6.9	6.0	6.5	7.8	10.9	13.1	14.5	14.8	13.3	10.4	8.5	6.7
	10%	DB	7.7	7.2	8.0	10.1	12.9	14.8	16.4	16.7	15.1	11.8	9.1	7.5
		MCWB	6.2	5.5	5.8	7.3	10.3	12.3	14.0	14.3	12.9	9.9	7.8	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	8.7	9.1	11.3	13.3	15.7	16.8	16.8	16.2	13.4	10.4	8.9
		MCDB	9.9	9.8	12.0	14.3	17.5	19.8	20.3	20.1	18.6	14.6	11.6	10.0
	2%	WB	8.1	7.2	7.9	9.4	12.2	14.4	15.7	15.9	14.8	12.0	9.5	7.9
		MCDB	9.0	8.4	9.8	11.5	14.8	17.1	17.9	18.2	16.5	12.9	10.6	9.1
	5%	WB	7.2	6.4	7.0	8.4	11.4	13.5	15.0	15.3	14.0	11.0	8.7	6.9
		MCDB	8.1	7.8	8.6	10.6	13.5	15.5	16.7	17.1	15.4	11.9	9.8	8.1
	10%	WB	6.4	5.7	6.2	7.6	10.6	12.7	14.3	14.6	13.3	10.2	7.8	6.1
		MCDB	7.4	7.1	7.7	9.6	12.5	14.3	15.9	16.3	14.6	11.4	9.0	7.4
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.4	5.5	6.0	5.7	5.0	4.6	5.0	5.4	5.1	4.7	4.4
		MCDBR	4.8	6.5	7.2	8.8	8.5	7.4	6.3	7.1	7.3	6.3	5.1	4.8
		MCWBR	3.5	4.2	4.4	5.1	4.6	4.0	3.5	4.0	4.3	4.1	3.7	3.3
	5% WB	MCDBR	4.2	5.3	6.2	7.3	7.1	6.6	5.9	6.3	6.0	4.8	4.6	4.3
		MCWBR	3.5	3.9	4.2	4.6	4.3	3.8	3.6	3.8	4.0	3.6	3.7	3.4
Clear-Sky Solar Irradiance	taub	0.301	0.308	0.334	0.369	0.360	0.380	0.366	0.363	0.355	0.336	0.310	0.290	
	taud	2.374	2.483	2.421	2.331	2.435	2.404	2.489	2.519	2.515	2.476	2.354	2.313	
	Ebn at Noon	660	799	847	858	887	869	875	859	818	747	632	581	
	Edh at Noon	54	68	89	111	105	110	100	91	81	68	54	47	
All-Sky Solar Radiation	RadAvg	0.51	1.33	2.37	3.81	4.88	5.15	4.67	4.04	2.66	1.46	0.68	0.41	
	RadStd	0.07	0.23	0.30	0.23	0.45	0.39	0.53	0.36	0.30	0.21	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	+0.41	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page